Sahaj Bhadja

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Summary

Software engineer completing an MSCS at USC with a strong **mathematics** and algorithms foundation and 2+ years building financial data systems, **SQL** databases, and **XML**/**CSV** extract-file workflows. Experienced in unit and integration testing, production support, technical documentation, and collaborative code reviews.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Data Structures, OOP, Operating Systems, Software Engineering

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built and modified software systems through **Python**/FastAPI development, **PostgreSQL** and SQL Server schemas, and ETL reconciliation pipelines processing **5K+ monthly jobs** at **99%+ reliability** for production data workflows.
- Conducted unit and integration testing with validation checks, prepared technical documentation, and supported production troubleshooting during peer code reviews and Agile standups across cross-database **SQL** transformation environments.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Programmed and debugged **Python**/Flask application logic with **MySQL**/AWS RDS backends, analyzing instructor-facing workflows, preparing design proposals, and delivering functional software through modular MVC architecture and REST APIs.
- Researched problems discovered during QA validation, implemented unit and integration tests on GitHub Actions, and conducted code reviews that reduced manual processing turnaround by **60%** within collaborative Agile delivery cycles.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Analyzed and implemented software modules processing **JSON**, **CSV**, and **XML** extract files through AWS Glue ETL pipelines, consolidating activity logs into normalized **SQL**-ready formats with automated reconciliation and error handling.
- Performed unit and integration testing with **20+ PyTest** suites in Jenkins CI/CD, researched QA-discovered defects, and participated in code reviews while providing production support across Linux, Windows, and macOS deployments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built and reviewed software systems on **Azure SQL** with optimized relational schemas syncing **2M+ records**, using Jenkins CI/CD and **Git**-based software configuration management to validate releases across billing modules.
- Conducted unit and integration testing through automated Jenkins pipelines, debugged deployment issues, and supported production releases while partnering with senior developers on systems analysis, programming activities, and technical documentation.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Developed **Python**/Flask software systems for **financial** transaction workflows, supporting multi-currency payment infrastructure, webhook automation, and structured data processing across **50K+** booking and payment records in production.
- Ingested flat-file and JSON extract data through Databricks analytics pipelines, prepared technical documentation for client delivery, and collaborated in code reviews while tuning **SQL** queries for sub-**200ms** API performance.

Skills

Software & Data: Python, software systems development, debugging, systems analysis, unit testing, integration testing, PyTest, production support, technical documentation, code reviews, problem research, Agile
Finance & Formats: SQL, Oracle (familiar), PostgreSQL, MySQL, Azure SQL, XML, CSV, flat files, JSON, ETL, AWS Glue, S3, financial industry, managed accounts (familiar), wealth management (familiar), mathematics, algorithms
Tools: Git, software configuration management, Jenkins, CI/CD, GitHub Actions, FastAPI, Flask, Docker, Linux, MATLAB (familiar), APL (familiar), Dyalog (familiar)

Projects

FinNexus Live | *TypeScript, Fintech, CI/CD, Financial Data*

- Crafted a fintech platform supporting **50K+ users** and **1M+ transactions** with Plaid bank linking, real-time account visibility, CI/CD automation, and security controls aligned with wealth management data standards.

PortfolioLab | *Python, SQL, CSV, XML, Portfolio Analytics*

- Built portfolio analytics tooling ingesting **CSV** and **XML** extract files into **SQL** schemas, enabling allocation reporting, reconciliation checks, and client-delivery workflows modeled on separately managed account data patterns.

Chat2SQL Code | *Python, SQL, Oracle-Style Queries, Git*

- Developed multi-database **SQL** tooling supporting PostgreSQL, MySQL, and Oracle-style query workflows with optimized relational schemas, Docker deployment, and **Git** version control for collaborative configuration management.